

Table I maps each major prior study to what it measured, what environment it used, and what it omitted, demonstrating the gap.

Study	What It Measured	Environment	Key Omissions
Bos et al.[6]	Kyber cycle counts at three security levels	C, assembly, x86-64	No classical baselines; no hash/AES; no hybrid; no distributions
Paquin et al.[7]	TLS 1.3 handshake latency under network variation	TLS stack, Open Quantum Safe	No isolated primitive timing; no hash; protocol conflates costs
Sosnowski et al.[8]	TLS 1.3 handshake with PQC algorithms	TLS 1.3, multi-algorithm	No hash/AES benchmarks; no hybrid decomposition; no distributions
Abbasi et al.[9]	ML-KEM across hardware platforms	ARM, x86-64, embedded C	No classical baselines; no hash/AES; no hybrid construction
Sonnino et al.[13]	Continuous benchmarking of production crypto library	C, production library, x86-64	Single library; not multi-family; no hybrid evaluation
Sarangi[12]	AES-128/192/256 performance across input sizes	Python environment	No key exchange or hash; not connected to PQC/hybrid context
Al-Safi & Shutyl[11]	SHA-2 vs SHA-3 throughput at varied input sizes	Software benchmark	No key exchange or AES; predates PQC; no hybrid consideration
Bhargavan et al.[39]	Formal verification of PQXDH hybrid key agreement	Symbolic analysis tools	No empirical performance data; no hash/AES/symmetric benchmarks
Turino et al.[40]	PQC, hybrid-PQC and classical benchmarks across x86 and ARM	C, liboqs, OpenSSL 3.5.0	No Python environment; no primitive-level distributions; no hybrid decomposition